Supplementary Information

Measurement resolution constrains fine-grained perturbation prediction

This Supplementary Information provides additional methodological detail and supporting analyses for the variant representations, prediction models and measurement-resolution framework used in the main text.

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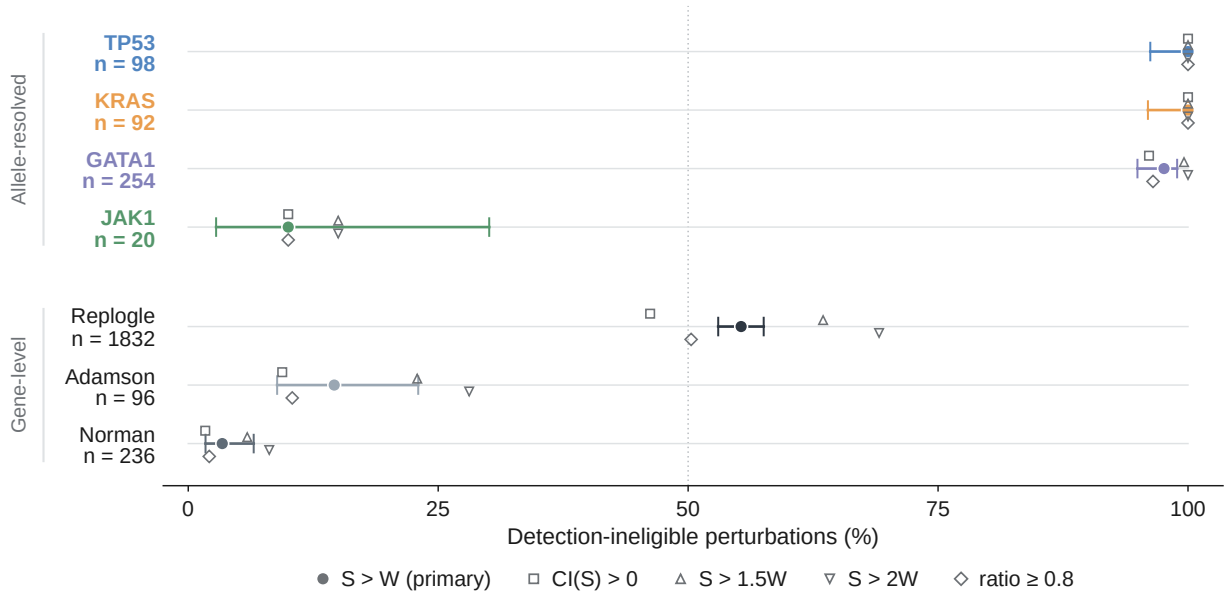
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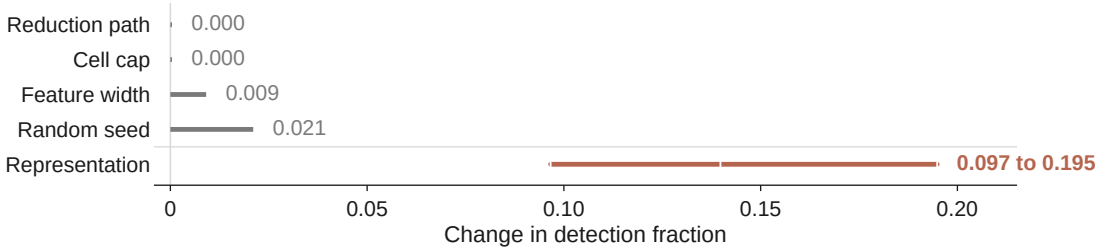
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Supplementary Figure 1: Detection regimes are stable across alternative decision criteria. Fraction of detection-ineligible perturbations in the four allele-resolved datasets and three gene-level Perturb-seq atlases under five criteria. The primary criterion compares perturbation-to-reference separation S with within-condition sampling variation W ($S > W$; filled circles). The alternatives (open symbols) apply a bootstrap lower bound on S, increased stringency ($S > 1.5W$ and $S > 2W$), or the distance ratio $D_{\text{self}}/D_{\text{null}} \geq 0.8$. Absolute fractions vary with the criterion, most for Replogle (46.2% to 69.1%), but the measurement regimes are unchanged under every criterion: TP53 and KRAS remain at the detection floor, GATA1 remains weakly resolved and JAK1 remains substantially separated from its reference. Whiskers, Wilson 95% confidence intervals on the primary criterion; symbols are offset vertically within a row for clarity; n, perturbations evaluable at native depth; dotted line, 50%.



Supplementary Figure 2: Detection fractions are stable to routine pipeline choices and shift with the expression representation. Bars, largest absolute change in the fraction of perturbations meeting the detection criterion when one analysis choice is varied on a fixed cohort: dimension-reduction path (exact, incremental and randomized; Norman), per-perturbation cell cap (capped versus uncapped; GSE306429 A375), feature width (five width-matched arms of Frangieh, Papalexi and PerturbMulti) and random seed (three seeds; GSE306429 A375). Representation, change from unnormalized counts to log-normalized expression of 2,000 highly variable genes; bar ends, Replogle and Norman; white tick, Adamson.

Supplementary Table 1: Structural and functional annotations used in the biophysical variant representation. Per-gene residue annotations for the position-dependent components of the feature vector θ. The fold-core term f_v was set to 1 within the listed structural cores and 0 otherwise. The functional-switch term s_v marks the listed switch residues. The hotspot/pathogenic term h_v was defined from external annotations without reference to the measured single-cell response.

Gene	Structural core ($f_v = 1$)	Functional switch ($s_v = 1$)	External functional or hotspot annotation ($h_v = 1$)
TP53	DNA-binding domain, residues 94–293	–	COSMIC/IARC missense hotspot codons 175, 176, 179, 220, 238, 245, 248, 249, 273, 282 and 285
KRAS	G-domain, residues 1–169	G12, G13 and Q61	COSMIC/IARC missense hotspot codons 12, 13, 59, 61, 117 and 146
GATA1	Zinc fingers ZF1 (204–228) and ZF2 (258–282)	Zinc-coordinating cysteines	Zinc-coordinating cysteines 204, 207, 225, 228, 258, 261, 279 and 282; ClinVar pathogenic or likely-pathogenic annotations
JAK1	Kinase domain, residues 866–1154	–	JH2 pseudokinase (583–855) and JH1 kinase (875–1153) functional modules

Supplementary Table 2: Prediction-model and measurement-resolution parameters. The six regression heads were crossed with the three primary molecular feature spaces (θ, ESM-1v and ESM-1v+θ) to form the 18 feature-based predictors evaluated in the main model grid. The lower panel lists the principal sampling parameters used for the measurement-resolution and repeated-pilot analyses.

Regression head	Family	Settings
Ridge regression	Linear (L2)	$\alpha = 1.0$
Lasso regression	Linear (L1)	$\alpha = 0.01$; maximum 2,000 iterations
Random forest	Tree ensemble	50 trees; maximum depth 6
Gradient-boosted trees	Tree ensemble	50 estimators; maximum depth 3; multi-output wrapper
k-nearest-neighbours	Neighbour	$k = 3$
Multilayer perceptron	Neural network	hidden layers 128 and 64; ReLU activation; maximum 500 iterations

Analysis	Representation, depths and partitions
Split-half depth ladder (Fig. 4c,d, Fig. 6f)	dataset-native analysis spaces for the allele-resolved arm: processed 939-gene count spaces for TP53 and KRAS, the log-normalized space of 2,477 highly variable genes for GATA1, and the log-normalized 2,000-gene space for JAK1. The Replogle, Norman and Adamson gene-level screens use library-size normalization to 10^4 counts per cell, log transformation and 2,000 highly variable genes. A 50-component PCA is fitted once per gene or dataset; 50 random cell splits are used at depths of 30, 50, 100 and 150 cells per half for the allele-resolved datasets and 25, 50, 100, 200, 400 and 800 for the gene-level datasets. A perturbation enters a rung at twice that rung's cells; allele-resolved pools are capped at 300 cells
Nearest-competitor and controlled-geometry sweep (Fig. 4e, Fig. 6d)	$m \in \{25, 50, 100\}$ cells per group with 10 random cell partitions; the measured configurations take four disjoint groups per variant and the constructed configurations five, the extra block supplying the direction that is dialled; constructed candidate-set sizes, including the target, of 20, 45 and 90 variants
Broader-panel criterion (Fig. 5)	unnormalized counts with no log transformation and no gene selection, 50-component PCA fitted once per resource-by-stratum dataset; 50 cells per group; 8 random cell partitions; 1,000 bootstrap resamples; seed 0; a perturbation enters at 200 cells
Repeated-pilot detection association (Fig. 6b)	50-component PCA, or truncated SVD for feature matrices too wide for exact decomposition, fitted before partitioning; 10 random pilot and evaluation partitions; 25- or 50-cell pilot scores averaged across partitions; detection eligibility defined from the median and percentile summaries of ten 100-cell-per-half evaluation draws; perturbed pilot and evaluation cells disjoint within a partition and reusable across partitions; reference draws independent and possibly overlapping; one AUROC after aggregation; 2,000 bootstrap resamples over perturbations
Split-half reproducibility reference	15 random split-half subsamples per variant, each half capped at 300 cells, scored with the same PDS evaluation as the predictors
Sensitivity representations	gene-expression space; pathway-score space where available
Minimum cells for pseudobulk prediction	5 per variant condition

Note. For expression spaces exceeding 200 genes, the random-forest and gradient-boosted models used $q = \min(50, G, n_{\text{train}} - 1)$ principal components of the training responses.

Supplementary Table 3: Released perturbation models and allele-conditioned adaptations under the unified evaluation. Except for the scVIDR control, PDS and Pearson-δ are evaluated on held-out protein-coding variants without using their response cells, with the same pseudobulk targets, scoring and distinct-variant aggregation as the feature-based predictors; PDS confidence intervals resample distinct variants. n, number of PDS queries.

Model family	Method	n	PDS (95% CI)	Pearson-δ
Response-blind allele-conditionable models				
Optimal-transport flow	CellFlow	303	0.493 (0.467–0.520)	0.16
Attribute disentanglement	Biolord	308	0.479 (0.453–0.505)	0.14
Latent arithmetic, allele-conditioned adaptation	scGen	308	0.474 (0.449–0.500)	0.20
Conditional-flow generative model	PerturbNet[†]	303	0.530 (0.504–0.555)	0.13
Feature regression	models in Fig. 2	305–308	0.471–0.489	0.58–0.67
Response-informed sensitivity control; excluded from response-blind inference				
Latent arithmetic, oracle-scaled adaptation	scVIDR[§]	308	0.486 (0.461–0.511)	0.18
Reference predictors				
Gene-level response	Gene-mean	308	0.478 (0.450–0.504)	0.67
No perturbation	WT-null	308	0.500	–
Models without an allele-resolved conditioning interface				
Foundation model	scGPT	–	n/a[‡]	–
Gene knowledge graph	GEARS	–	n/a[‡]	–
Virtual-cell model	STATE	–	n/a[‡]	–

[†]Run across all five scored within-gene settings (492 gene–setting–variant evaluations, 303 distinct variants) and scored in its 50-dimensional output representation; task-matched permutation null mean 0.515, one-sided $P = 0.070$ (Methods).

[‡]scGPT, GEARS and STATE condition on perturbed-gene identity, so all variants of a gene receive the same prediction.

[§]Response-informed: for GATA1 the scaling variable is the measured variant-to-WT expression distance normalized over the full GATA1 cohort (Supplementary Note 3). The variable is missing for JAK1, so the 26 JAK1 queries receive the no-ranking value of 0.5; finite prediction coverage and the Pearson-δ denominator are 282 variants.

Supplementary Table 4: Measurement resolution and prediction performance across the four allele-resolved datasets. The split-half ratio $D_{\text{self}}/D_{\text{null}}$ compares within-variant sampling variation with variant-to-reference separation. The reproducibility reference measures full-pool allele identification between disjoint cell samples using the same PDS framework as the predictors, aggregated over distinct variants; n, distinct variants scored. These are the same values used in Fig. 3. Sibling pairs resolvable gives the fraction of same-gene variant pairs separated beyond the mean of their two split-half distances, and nearest-sibling identifiable the stricter fraction of variants whose closest sibling remains separated beyond that variant's own split-half distance. Detection eligibility gives the fraction of variants evaluable at native depth (JAK1, 20 of 26) that are distinguishable from the corresponding reference.

Gene	$D_{\text{self}}/D_{\text{null}}$	Reproducibility reference (95% CI), n	Sibling pairs resolvable	Nearest-sibling identifiable	Detection-eligible	Best feature-based PDS
TP53	0.96	0.472 (0.441–0.505), 59	0%	0%	0%	0.479
KRAS	1.00	0.481 (0.431–0.533), 51	0%	0%	0%	0.505
GATA1	0.88	0.578 (0.552–0.604), 172	3.8%	0%	2.4%	0.509
JAK1	0.21	0.807 (0.729–0.880), 26	78.5%	15.4%	90%	0.448

Note. Native depth denotes the deepest evaluated split-half rung supported by each variant, capped at 150 cells per half for these datasets. The final column reports the highest PDS among the feature-regression models for each gene under the primary scoring. Released perturbation models are reported separately in Supplementary Table 3.

Supplementary Table 5: Three distinct measurement resolutions across six prespecified perturbation screens. The three quantities are different biological comparisons: detection is separation from the reference condition, nearest-competitor identification is separation from the closest competing perturbation, and the reproducibility reference and $P(\text{order})$ describe benchmark-level ranking resolution, the first as split-half perturbation ranking and the second as the bootstrap probability of recovering the design ordering of the controlled predictor family. A screen can support one comparison and not another.

Dataset	n	Detection (%)	Nearest-competitor identification (%)	Reproducibility reference	$P(\text{order})$
McFarland 2020	10	90.0	20.0	0.928	0.418
Tahoe-100M	7	71.4	42.9	0.929	0.106
Norman 2019	195	68.7	0.5	0.898	0.997
Replogle 2022	414	51.2	1.4	0.865	0.959
Adamson 2016[†]	93	48.4	3.2	0.853	0.977
Virtual Cell Challenge	139	25.9	4.3	0.806	0.903

Note. The operating points were $> 50\%$ of perturbations detected, $> 50\%$ identified against the nearest competitor and a split-half reproducibility reference > 0.65 (Methods).

[†]Cells without a perturbation assignment were excluded, leaving 93 scored perturbations.

Supplementary Table 6: Public datasets assessed for inclusion in the primary allele-resolved benchmark. Datasets were assessed against three operational requirements: **R1**, an accessible genotype-linked single-cell expression representation; **R2**, sufficient sibling coding variants of the same gene, each with at least five assigned cells, for within-gene held-out evaluation; and **R3**, compatibility with a common held-out-variant prediction protocol. The table lists the public datasets assessed for this study.

Study (accession)	Variants assayed	R1	R2	R3	Outcome
Ursu et al. 2022 (GSE161824)	*TP53* 98; *KRAS* 92	yes	yes	yes	included
Martin-Rufino et al. 2023 (GSE215253)	*GATA1* 254	yes	yes	yes	included
Cooper et al. 2024 (PRJEB48915)	*JAK1* 26	yes	yes	yes	included
Kim et al. 2024, TISCC-seq (PRJNA880341)	*TP53*, > 100	no	–	–	no released cell-to-genotype expression mapping
Hollek et al. 2026 (GSE299651)	50 transgenes across five cell lines	yes	no	–	insufficient sibling alleles per gene
Xu et al. 2023, preprint	*CDKN2A*, *TP53*, *SOD1*	no	–	–	no accessible expression dataset identified

Note. Dataset availability was assessed on 11 August 2026. R1 refers to released genotype-linked expression products.

Supplementary Note 1: Robustness of measurement-resolution estimates

Distance metric and expression representation. The detection regime held across five distance families and five detection criteria (Supplementary Fig. 1). At 50 cells per half in the canonical PCA space, the ratio $D_{\text{self}}/D_{\text{null}}$ remained near one for TP53, KRAS and GATA1 and substantially lower for JAK1 across energy distance, Gaussian-kernel maximum mean discrepancy, sliced-Wasserstein distance, cosine distance and Euclidean distance. The corresponding ranges were 1.01–1.03 for TP53, 0.95–0.99 for KRAS, 0.90–1.04 for GATA1 and 0.05–0.40 for JAK1.

Repeating the native-depth analysis directly in gene-expression space gave the same dataset-level interpretation. Detection eligibility was 0% for TP53 and KRAS, 1.2% for GATA1 and 90%

for JAK1, compared with 0%, 0%, 2.4% and 90%, respectively, in the primary PCA representation. Pathway-score representations, available for GATA1 and the three gene-level Perturb-seq datasets, generally produced lower absolute eligibility fractions and preserved the measurement regimes.

Batch-disjoint split-half sensitivity analysis. The split-half reference draws both groups from a pooled shuffle, so both carry the same batch composition. For datasets with per-cell experimental batch information, we replaced the pooled shuffle with two disjoint batch groups at matched depth. At 50 cells per group the dataset-level reference moved by at most 0.038 PDS (largest in sci-Plex, 0.619 versus 0.581; smallest in Replogle, 0.789 versus 0.784). TP53 and KRAS remained at chance under both definitions, and the dataset-level regimes were preserved throughout. JAK1 was sequenced in a single lane and has no batch split.

Reference-population variability. We tested whether weak perturbation-to-reference separation could instead arise from heterogeneity of the reference population. At matched sampling depth, we compared within-variant split-half distance D_{self}, variant-to-reference distance D_{null} and the distance between independent reference samples D_{wtwt}. For TP53, KRAS and GATA1, $D_{\text{wtwt}}/D_{\text{null}}$ ranged from 0.91 to 1.09 at every depth their reference cells allowed (50 cells per half for TP53, up to 150 for KRAS and 300 for GATA1), indicating that variant-to-reference separation was comparable to variation between independent reference samples. In JAK1, by contrast, D_{null} exceeded D_{wtwt} by 8.4-fold at 50 cells per half (14 variants), 31-fold at 150 (6 variants) and 7.4-fold at 300 (1 variant). The weak detection regime in the first three datasets therefore reflects limited perturbation separation relative to sampling variation rather than an unusually heterogeneous reference population.

Supplementary Note 2: Robustness of allele-prediction conclusions

Dependence among variants at the same residue. Several benchmark residues contain more than one amino-acid substitution, so variant-level observations are not completely independent. We therefore repeated the primary uncertainty analysis by resampling residue-level clusters, 224 for the 308 variants, rather than individual variants. Twenty-three of the 24 predictors with confidence intervals overlapped the analytic full-pool PDS expectation of 0.5, with PerturbNet remaining the sole exception by that descriptive comparison, and Pearson-δ remained positive for every feature-based regression model. For PerturbNet, the task-matched restricted-label permutation test gives $P = 0.070$ (Methods). The separation between direction recovery and full-pool allele identification is therefore unchanged when substitutions at the same residue are treated as a single resampling cluster.

Dependence among resolved sibling pairs. Because resolved pairs in Fig. 3e can share a variant, we also performed a variant-node bootstrap on the fixed cross-fitted resolved-pair cohorts. GATA1 contained 425 edges incident to 131 variants and JAK1 contained 116 edges incident to 21 variants. In each of 1,000,000 draws, the n incident variants were sampled with replacement, an original edge (i, j) received weight $c_i c_j$ from the sampled endpoint multiplicities, and its pair score entered a weighted mean. The same node draw was applied to the measurement, each model and their paired difference. Repeated copies of one original variant did not create an unobserved self-edge. For GATA1, the split-half measurement accuracy was 0.988 (95% node-bootstrap interval, 0.972–0.998) and the highest response-blind point estimate, Ridge-ESM, was 0.780 (0.603–0.901), with a paired model-minus-measurement difference of -0.208 (-0.382 to -0.093). For JAK1, the corresponding values were 1.000 (1.000–1.000) for the measurement and 0.762 (0.594–0.883) for Biolord, with a difference of -0.238 (-0.406 to -0.117). The response-informed scVIDR sensitivity control had the largest displayed GATA1 point estimate, 0.830 (0.634–0.937), and a difference of

-0.159 (-0.356 to -0.050). Every fitted predictor's paired interval remained below the split-half measurement. These intervals are conditional on the resolved-pair selection and cell partitions and are not adjusted for selecting the highest model.

Alternative molecular representations. We evaluated the same held-out-variant task using the biophysical representation θ, ESM-1v, four ESM2-derived representations and AlphaMissense. The ESM2 representations were built from ESM2-650M (`esm2_t33_650M_UR50D`) final-layer per-residue hidden states. For each variant the mutant sequence was formed by applying every listed substitution to the reference sequence, with each substitution matched to the wild-type residue at that position. Sequences exceeding the model's 1,022-residue limit, which applies to JAK1 at 1,154 residues, were cropped to a 1,022-residue window centred on the mutated positions, and the wild-type sequence was cropped identically so that every difference is taken between matched windows. Writing $\mathbf{h}^{\mathrm{mut}}$ and $\mathbf{h}^{\mathrm{wt}}$ for the two hidden-state matrices and p for a mutated position, the four representations are the mutant global embedding, the residue mean of $\mathbf{h}^{\mathrm{mut}}$; the global difference, the residue mean of $\mathbf{h}^{\mathrm{mut}}$ minus that of $\mathbf{h}^{\mathrm{wt}}$; the mutated-residue difference, $\mathbf{h}^{\mathrm{mut}}_p - \mathbf{h}^{\mathrm{wt}}_p$; and the local-window difference, the mean of $\mathbf{h}^{\mathrm{mut}} - \mathbf{h}^{\mathrm{wt}}$ over residues $p \pm 16$. For variants carrying more than one substitution, the last two are averaged over the mutated positions. AlphaMissense entered as its per-variant pathogenicity score, matched on wild-type residue, position and mutant residue; variants carrying more than one substitution have no such entry and were excluded from that representation. Across these representations, full-pool allele identification remained near the null while transcriptional direction remained recoverable. Representation coverage differed because some molecular annotations were unavailable for individual variants; AlphaMissense scores were available for 374 of the 470 benchmark variants, compared with 470 for θ, 467 for ESM-1v and the corresponding available sets for the ESM2 representations. These analyses test whether the principal result depends on the molecular encoding.

Numerical convergence of Lasso-ESM to the gene-constant reference. For Lasso-ESM, held-out predictions in TP53, KRAS and GATA1 converged numerically to the Gene-mean reference, agreeing to a maximum relative difference of 4×10^{-7}, which is the rounding scale of the single-precision prediction arrays; JAK1 did not show this equivalence (maximum relative difference 7.4). One of the 18 feature-based predictors therefore coincides with the gene-constant reference in three of the four datasets. Lasso-θ and Lasso-ESM+θ did not converge in this way (maximum absolute differences 6×10^{-2} to 9×10^{-2}).

Residual-axis permutation calibration. Residual-axis PDS was compared with its task-matched restricted-label permutation distribution (Methods). Permutations were run in two stages: 50 permutations for every method, followed by 1,000 permutations for the methods whose preliminary permutation P fell below 0.05. Residual-axis P values were then Bonferroni-adjusted across the 25 scored methods. Unadjusted values were 0.0020 for scGen and 0.0080 for Biolord, giving adjusted values of 0.050 and 0.200. With 1,000 permutations P moves in steps of $1/1001$, so the adjusted scGen value of 0.050 cannot be separated from the 0.05 threshold and is reported as unresolved.

Evaluation setting, scoring geometry and standardization. The separation between direction recovery and full-pool allele identification held under every evaluation setting, scoring geometry and standardization convention tested.

The same predictors were scored under each evaluation setting separately. Averaged over the 19 non-null predictors, mean PDS and mean Pearson-δ were 0.481 and 0.667 under random hold-out, 0.494 and 0.655 under positional extrapolation, 0.517 and 0.511 under mechanistic extrapolation, 0.496 and 0.664 under the compatibility protocol, and 0.431 and 0.575 under low-depth evaluation. No predictor's PDS confidence interval lay above chance in any setting. Low-depth evaluation weakened both axes rather than revealing identification, and placed the intervals of 6

of the 19 predictors below chance, which is expected when the measured targets themselves are the noisiest: a prediction can then sit closer to a better-sampled competitor than to its own target.

Scoring geometry was varied by recomputing the identical predictions and targets under L1 and L2 distances alongside the cosine distance used throughout. The per-head mean PDS moved by at most 0.017 under L1 and under L2, the largest shift in each case belonging to the same head, so no predictor crossed from chance to reliable identification under a different geometry.

Finally, replacing the training-only per-gene standardization used in the primary analysis with all-cell standardization raised scores for 18 of 20 predictors on PDS and 19 of 20 on Pearson-δ, by at most 0.033 PDS and 0.030 Pearson-δ, both for the same gradient-boosted head. The training-only convention used throughout is therefore the more conservative.

Training-response subspace control in JAK1. The JAK1 dataset provides the clearest setting in which the measurement itself contains reproducible allele-level information but the evaluated predictors remain close to chance. Projecting each held-out JAK1 response onto the span of the measured training-variant responses reached PDS 0.690 (95% CI, 0.638–0.742), against 0.482 for the best feature-based predictor (MLP-ESM) and 0.792 for the split-half reproducibility reference, each averaged over the same four cell splits; under the distinct-variant aggregation of Supplementary Table 4 the latter two are 0.448 and 0.807. The span of measured training-variant responses therefore contains substantially more allele-discriminative structure than the fitted feature-based models recover. Because the projection uses the held-out measured response, it is a reference for that subspace rather than a prediction.

Exact identification remained substantially more difficult than ranking-level discrimination: the projection achieved top-1 accuracy 0.049, whereas the split-half measurement reference reached 0.247 over the same four cell splits.

Supplementary Note 3: Implementation of the released perturbation models

All scored released-framework implementations were run on the same arrays, cell caps and variant partitions as the feature-based predictors: per-gene standardized expression matrices, at most 300 cells sampled without replacement per variant condition under a fixed per-gene seed, and the five scored evaluation settings. CellFlow and PerturbNet were scored on 303 distinct variants (492 gene–setting–variant evaluations), and the other models on all 308. Every model predicts a pseudobulk response, formed as the mean of its predicted profiles over the reference cells minus the measured reference mean, so that all scored entries of Supplementary Table 3 are scored with the same PDS evaluation. Held-out variants contributed no cells to any parameter fit. Except for the scVIDR control described below, held-out response cells also contributed no information to prediction-time conditioning or scaling.

scGen and the response-informed scVIDR sensitivity control. Both were implemented with one latent-arithmetic pipeline and differ in how the latent shift is scaled. A variational autoencoder with encoder $d \to 256 \to 256$, a 32-dimensional latent space and a mirrored decoder was trained per gene and per split on the training-variant cells together with the reference cells, minimizing mean squared reconstruction error plus a Kullback–Leibler term weighted at 0.01, with Adam at learning rate 10^{-3}, batch size 512, 120 epochs, no early stopping and a fixed seed. For each training variant the latent shift was the difference between the mean latent code of that variant's cells and the mean latent code of the reference cells. A ridge regression with $\alpha = 1$ was then fitted from the six-dimensional biophysical representation θ to those latent shifts, which is what makes the adaptation allele-conditioned: the original methods index a categorical perturbation, whereas here

the shift is a continuous function of molecular variant features and is therefore defined for a variant never seen in training. For a held-out variant the predicted shift was added to the latent code of every reference cell and decoded. **scGen** uses that shift unchanged and is the response-blind entry in the main comparison.

scVIDR additionally rescales the fitted shift by $\max(r_v^{\text{scale}}, 0.05) / \text{mean}_{u \in \text{train}} |r_u^{\text{scale}}|$. The provenance of this model-specific scalar differs by gene. For TP53 and KRAS, r_v^{scale} is the AlphaMissense pathogenicity score, with synonymous or wild-type entries set to zero; these are response-independent external attributes. For GATA1, however, the stored value is

$$r_v^{\text{scale}} = \frac{\|\bar{X}_v - \bar{X}_{\text{WT}}\|_2}{\max_w \|\bar{X}_w - \bar{X}_{\text{WT}}\|_2}, \tag{S1}$$

where the means are calculated from the measured GATA1 expression matrix and the maximum uses the full GATA1 cohort. Thus a held-out GATA1 variant's response cells and a full-cohort response statistic enter its scaling at prediction time, for all 172 distinct scored GATA1 variants. Accordingly, scVIDR is reported as an oracle-scaled, response-informed sensitivity control. The r^{scale} field is unpopulated for every JAK1 variant, so the scaling factor is non-finite and the implementation returns no finite JAK1 prediction. The displayed PDS therefore contains 308 queries, with the 26 JAK1 queries assigned the no-ranking value of 0.5, whereas finite prediction coverage and the Pearson-δ denominator are 282 variants.

Biolord. Biolord was run through its released implementation with the six θ components registered as ordered attributes and no categorical attributes, so the reference cells enter as the all-zero attribute vector. One model was trained per gene and split with a 32-dimensional latent space, a decoder of width 256 and depth 2, attribute networks of width 128 and depth 2, a 16-dimensional ordered-attribute latent, reconstruction penalty 10^4, unknown-attribute penalty 10 and unknown-attribute noise 10^{-1}; training used 100 epochs at batch size 512 with a cosine schedule, learning rate 10^{-3} for the latent, decoder and attribute networks, and a random 10% of cells held out as a validation split. No classifier heads were trained and early stopping was disabled, so the reported model is the final epoch rather than a selected checkpoint. Prediction for a held-out variant set that variant's θ on every reference cell and read the model's predicted expression.

CellFlow. CellFlow was run as an optimal-transport flow-matching model on the same arrays, with the perturbation covariate declared as a single categorical variant column whose representation is the variant's 1,280-dimensional ESM-1v embedding, and with the reference cells labelled as the control condition carrying a zero embedding. Held-out variants' ESM-1v embeddings, but not their cells, are available at prediction time, which lets the trained model condition on a variant it never saw in training. The condition embedding had dimension 64, the condition encoder hidden dimensions were 512, 512, 64 with mean pooling, and the decoder dimensions were 256, 256; training ran for 8,000 iterations at batch size 256. Prediction pushed the reference cells forward under the held-out variant's condition.

PerturbNet. PerturbNet was evaluated using ESM-1v conditioning under all five scored within-gene settings: random holdout, positional extrapolation, mechanistic extrapolation, low-depth evaluation and compatibility. The respective numbers of gene–setting–variant evaluations were 83, 229, 51, 59 and 70, for 492 evaluations in total. Because variants can appear in more than one setting, these collapse to 303 distinct gene–variant units: 59 TP53, 51 KRAS, 172 GATA1 and 21 JAK1 variants. Its output representation, aggregation and task-matched permutation test are given in Methods.

CPA. The CPA adaptation (`cpa-tools` 0.8.8 with scvi-tools 0.20.3) replaced the categorical perturbation representation with the continuous six-dimensional θ encoding, leaving the out-of-

distribution conditioning path otherwise intact. Its adversarial autoencoder did not converge (non-finite loss), so CPA has no prediction scores.

Models without an allele-resolved conditioning interface. scGPT, GEARS and STATE condition on perturbed-gene identity through a gene token, a knowledge-graph node and a one-hot perturbation identity respectively. None can ingest a held-out variant's molecular features, and all variants of a gene collapse onto a single prediction, so the 470 variant conditions reduce to four conditioning keys.

Supplementary Note 4: Response geometry and separation statistics

Substrate and noise. The configurations compared in Fig. 6d are built on measured TP53 and KRAS cells rather than on simulated expression. Across the depths examined, the between-variant signal of these two datasets satisfies $|\rho^2| \leq 0.005$, where ρ^2 is the noise-corrected mean pairwise separation defined in Methods, with every interval covering zero, so they supply realistic cell-sampling noise without contributing allele-specific signal of their own. The construction adds signal to these measured cells while leaving their sampling noise unchanged, so the signal-to-noise ratio is set directly.

Cell partition. Each variant's cells are divided, from a single shuffle, into five disjoint groups of m cells: a direction block, a query block Q, a target block T, and two signal blocks S_1 and S_2. Q and T each subtract their own wild-type half-mean, matching the convention used for the empirical split-half reference, while S_1 and S_2 share one, which therefore cancels from their difference. The reproducibility reference is measured on Q and T and the separation statistics on S_1 and S_2, so the two axes of every point come from different cells. The direction block enters no score, so the injected directions are independent of the noise they are added to.

Constructed separation. Directions are the mean profile of each variant's direction block, centred across variants so that the injected signal contains no component shared across all variants, and normalized to unit length. Their geometry is then reshaped in two ways: the direction set is truncated to a rank $r \in \{\text{full}, 5, 3, 2, 1\}$ by singular value decomposition, and each variant's direction u_v is multiplied by an independent lognormal amplitude a_v with $\log a_v \sim \mathcal{N}(0, s^2)$, $s \in \{0, 1, 2\}$, so $s = 0$ leaves the amplitudes equal. The reshaped set is then rescaled so that its mean pairwise squared separation equals exactly 2, which fixes the mean while leaving the geometry free, and multiplied by $a = \sqrt{\tau\, \hat{\eta}_0^2}$, where $\hat{\eta}_0^2$ is the substrate's own per-profile noise measured before any signal is added and $\tau \in \{0, 0.05, 0.2, 0.6, 2, 6\}$ is the target. The resulting mean pairwise squared separation is $2\tau\hat{\eta}_0^2$, so each configuration has $\rho^2 = \tau$ by construction. The same vector is added to Q, T, S_1 and S_2.

Factor grid. Two substrates, three sampling depths (25, 50 and 100 cells per group), three candidate-set sizes including the target (20, 45 and 90 variants), six signal levels, five ranks and three amplitude dispersions define 1,620 configurations. Neither substrate contains 90 variants with five disjoint groups of 100 cells, so 180 configurations are unavailable and 1,440 configurations remain. Each configuration is scored with the same tie-aware PDS evaluation used for every model and for the empirical split-half reference, comparing Q against T, and averaged over ten random cell splits.

Normalized separation metrics. At depth m, each variant v contributes two profiles $S_{v,1}, S_{v,2} \in \mathbb{R}^d$ from disjoint groups of m cells, each a cell mean minus a shared reference half-mean; in the constructed configurations these are the signal blocks. With $H = 2$ blocks per variant and n variants,

the per-profile sampling noise is

$$\hat{\eta}^2 = \frac{1}{n}\sum_{v=1}^{n}\frac{1}{H-1}\sum_{h=1}^{H}\left\|S_{v,h}-\bar{S}_v\right\|^2, \qquad \bar{S}_v = \frac{1}{H}\sum_h S_{v,h}. \tag{S2}$$

Mean pairwise separation subtracts the noise carried by a distance between two block means,

$$\widehat{\text{pair}}^2 = \binom{n}{2}^{-1}\sum_{i<j}\left\|\bar{S}_i-\bar{S}_j\right\|^2, \qquad \hat{\delta}^2 = \widehat{\text{pair}}^2 - \frac{2\hat{\eta}^2}{H}, \qquad \rho^2 = \frac{\hat{\delta}^2}{2\hat{\eta}^2}. \tag{S3}$$

Nearest-competitor separation replaces the mean over pairs with each variant's closest competitor, selected on one block and evaluated on the other, symmetrized over the two directions:

$$j_v^{(1)} = \arg\min_{u\neq v}\left\|S_{v,1}-S_{u,1}\right\|^2, \qquad j_v^{(2)} = \arg\min_{u\neq v}\left\|S_{v,2}-S_{u,2}\right\|^2, \tag{S4}$$

$$s_v = \tfrac{1}{2}\left(\left\|S_{v,2}-S_{j_v^{(1)},2}\right\|^2 + \left\|S_{v,1}-S_{j_v^{(2)},1}\right\|^2\right) - 2\hat{\eta}^2. \tag{S5}$$

The subtracted term is $2\hat{\eta}^2$ rather than $2\hat{\eta}^2/H$ because both distances are taken between single measurements rather than between block means. Under independent, mean-zero and exchangeable per-profile sampling errors with a common variance across variants, s_v then has expectation zero when no between-variant signal is present; with variant-specific variances, one pooled noise term need not equal the expected distance noise for every selected pair. Separations were computed within each of $S = 10$ cell partitions against the pooled $\hat{\eta}^2$, averaged across partitions, and only then summarized,

$$\rho^2_{\text{nn}} = \frac{\text{median}_v\left(\frac{1}{S}\sum_{s=1}^{S} s_v^{(s)}\right)}{2\hat{\eta}^2}, \tag{S6}$$

in that order, because separations formed from partition-averaged profiles would carry only $1/S$ of the noise the correction removes and would be over-corrected. Since the noise expectation has already been subtracted, $\rho^2_{\text{nn}} = 0$ means that nearest-competitor separation equals what sampling noise alone produces. Noise correction can send an individual separation below zero near the measurement floor; the median tolerates this, whereas a geometric mean does not and is reported as undefined rather than floored. Intervals are 95% percentile bootstraps of the median over the n variants, with 2,000 resamples and a fixed seed, holding $\hat{\eta}^2$ and the cell partitions at their point estimates.

Uniform candidate subsampling. Let the full pool contain $n-1$ competitors of which l are strictly closer to the prediction than the true target and $e-1$ are tied with it. Drawing $K-1$ competitors uniformly without replacement gives

$$E[L_K] = (K-1)\frac{l}{n-1}, \qquad E[E_K] = (K-1)\frac{e-1}{n-1}, \tag{S7}$$

and since $\text{PDS}_K = 1-(L_K + E_K/2)/(K-1)$,

$$E[\text{PDS}_K] = \text{PDS}_n. \tag{S8}$$

Reducing the candidate set uniformly therefore changes the resolution of the percentile, not its expectation (Fig. 6c).

Scope of the reported correlations. The Spearman correlations quoted in Fig. 6d are computed over all 1,452 plotted points, that is the 1,440 constructed configurations together with the twelve measured gene and depth conditions. Restricted to the 1,440 constructed configurations alone they are 0.763 for mean separation and 0.961 for nearest-competitor separation.

Supplementary Note 5: Broader-panel sensitivity and the sampling-depth projection

Sensitivity of the external perturbation panel. Routine analysis choices, namely the dimension-reduction path, per-perturbation cell cap, feature width and random seed, moved the panel's detection fraction by at most 0.021 and left every evaluated configuration on the same side of the 50% detection operating point. The expression representation moved it by 0.097 to 0.195, and the count-space criterion used throughout Fig. 5 returned the lower fraction on all three resources evaluated under both pipelines (Supplementary Fig. 2).

The sampling-depth projection. The projection reported in Fig. 6e inverts the controlled-geometry sweep of Supplementary Note 4.

Choice of axis. Four summaries of measured response geometry were scored by Spearman correlation with the split-half reproducibility reference over all 1,452 constructed and measured configurations: mean pairwise separation ρ^2 (0.767), median nearest-competitor separation ρ^2_{nn} (0.961), its geometric-mean counterpart (0.818) and the fraction of perturbations whose nearest competitor exceeds the noise (0.950). The median nearest-competitor separation was taken as the axis.

Monotonicity screen. Three quantities were candidates for inversion: the split-half reproducibility reference, the probability of recovering the complete design-defined ordering of the controlled predictor family, and the probability of selecting its best member. Each was scored by its Spearman correlation with the axis restricted to configurations above the median axis value. Only the reproducibility reference was retained ($+0.918$). Both recovery probabilities turn over in that range (-0.280 and -0.248, with the turnover near $\rho^2_{\mathrm{nn}} = 0.043$, at a reference of about 0.705), so a monotone calibration fitted to either would return incorrect depths in the high-resolution regime.

Calibration. Configurations were stratified by candidate-set size into $n \leq 30$, $30 < n \leq 70$ and $n > 70$, holding 543, 541 and 368 configurations, a stratum being dropped if it held fewer than 20. No functional form was assumed within a stratum. The axis was cut at its empirical quantiles at levels $0, 1/24, \ldots, 1$; each bin holding at least three configurations contributed a knot at the median axis value and the median reference of that bin; and the knot ordinates were then replaced by their running maximum. All three strata yielded 24 knots, reaching maximum references of 1.000, 1.000 and 0.997 respectively. The strata contain the constructed configurations together with the twelve measured gene-by-depth configurations, so the projections reported for those datasets are in-sample with respect to the calibration.

Inversion. For a target τ the required axis value $x^*(n, \tau)$ is the smallest knot abscissa whose monotonized ordinate reaches τ, with no interpolation between knots, and the depth projected from a pilot at m_0 cells per group is $\hat{m}(\tau) = m_0\, x^*(n, \tau)/\rho^2_{\mathrm{nn}}(m_0)$. No depth is returned when the stratum held fewer than 20 configurations, when the calibration never reaches τ, when the pilot estimate is not positive, or when the pilot's lower confidence bound does not exceed zero. Applied to the four datasets at 25, 50 and 100 cells per group and at $\tau \in \{0.7, 0.9\}$, the projection returned a finite depth for 8 of 24 combinations: all six for JAK1 (10.4 and 55.1 cells per group from the 25-cell pilot, 12.2 and 64.9 from the 50-cell pilot, 18.6 and 98.9 from the 100-cell pilot) and two for GATA1 (62.1 and 231.5, both from the 25-cell pilot), and none for TP53 or KRAS. In every combination without a projection, the pilot's lower confidence bound did not exceed zero; the first two conditions never applied.

Domain and uncertainty. The constructed configurations span candidate-set sizes of 20, 45 and 90, whereas the eight finite projections were read off at 247 for GATA1 and at 14, 9 and 5 for JAK1, so every reported projection extrapolates in candidate-set size. No interval is placed on a projected depth: the pilot lower bound acts as a condition and is not propagated.